Spotify Songs Analysis

Data Exploration & Dashboard Development

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1. Introduction

Dataset

Dataset title	Most Streamed Spotify Songs 2023
Source	Kaggle
Format	CSV (spotify-2023.csv)
Size	953 songs and 24 columns

Description: The dataset contains the most streamed songs on Spotify in 2023. These are real songs by real artists. Each row is one song and includes the song name, artist, release date, number of streams, how many playlists and charts it is in (Spotify, Apple Music, Deezer and Shazam), and music features like danceability, energy, valence (happiness) and bpm (speed).

Why I selected it: The data is real and the topic is fun and easy to understand. The data also needs real cleaning (numbers saved as text, missing values, duplicates and a wrong value), and it has both numeric and categorical columns for the charts and the dashboard filters.

Problem Definition

What is the dataset about? Which songs and artists were the most popular on Spotify in 2023, and what makes a song popular.

Main questions:

- Q1: Which artist has the most songs in the list?
- Q2: What are the most streamed songs?
- Q3: Are danceable songs streamed more?
- Q4: Are newer songs more common in the list?
- Q5: Are there more Major or Minor songs?

Expected insights: I expected famous artists like Taylor Swift to have many songs, new songs to be the most common, and danceable songs to have more streams.

2. Data Exploration

The data was loaded with Pandas using `encoding='latin-1'` because some song names have special characters. Then it was checked with `head()`, `shape`, `info()`, `describe()` and `isnull().sum()`.

Main Columns

Column	Description	Type when loaded
track_name	Song name	object
artist(s)_name	Artist name(s)	object
released_year / month / day	Release date	int64
streams	Total number of streams on Spotify	object (should be a number)
in_spotify_playlists	Number of Spotify playlists	int64
in_deezer_playlists	Number of Deezer playlists	object (should be a number)
in_shazam_charts	Rank in Shazam charts	object (should be a number)
bpm	Speed of the song	int64
key / mode	Musical key and Major/Minor	object
danceability_%, energy_%, valence_% ...	Music features from 0 to 100	int64

Problems Found

Problem	Column	Count
Text instead of a number (wrong value)	streams	1 song
Numbers with commas saved as text	in_deezer_playlists	79 values
Numbers with commas saved as text	in_shazam_charts	7 values
Missing values	in_shazam_charts	50
Missing values	key	95
Duplicate songs (same name and artist)	track_name, artist(s)_name	4

3. Data Cleaning

Step	What was done
Convert streams	Converted streams to numbers. One song had text instead of a number, so it was dropped.
Convert playlists / charts	Removed the commas from in_deezer_playlists and in_shazam_charts, then converted them to numbers.
Missing in_shazam_charts	Filled with 0, because a missing value means the song is not in the Shazam charts.
Missing key	Filled with "Unknown". We cannot know the key, and dropping 95 songs (10%) would lose a lot of data.
Remove duplicates	Removed 4 songs that appeared twice with the same name and artist.
Rename columns	Renamed columns to simple names (artist(s)_name to artist_name, danceability_% to danceability ...).
Save	Saved the result as clean_data.csv.

Result

	Before cleaning	After cleaning
Rows (songs)	953	948
Missing values	145	0
Numeric columns saved as text	3	0
Duplicate songs	4	0

Only 5 songs were removed (1 wrong value and 4 duplicates), so almost all the data was kept.

4. Exploratory Data Analysis (EDA)

Univariate Analysis

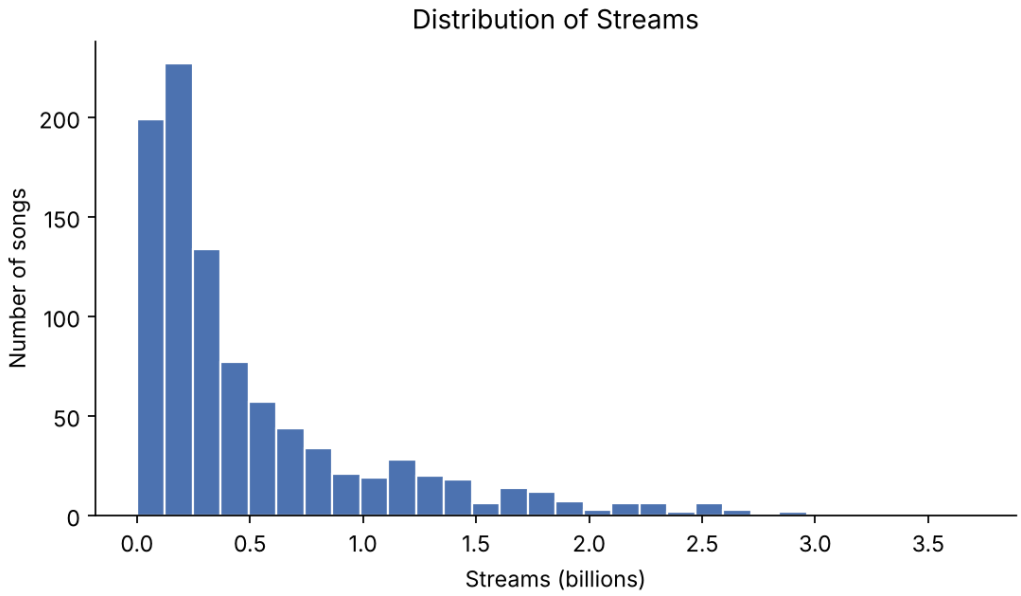


Figure 1: Histogram of streams. Most songs have less than 1 billion streams. The data is right-skewed: the median is about 288 million and the mean is about 514 million.

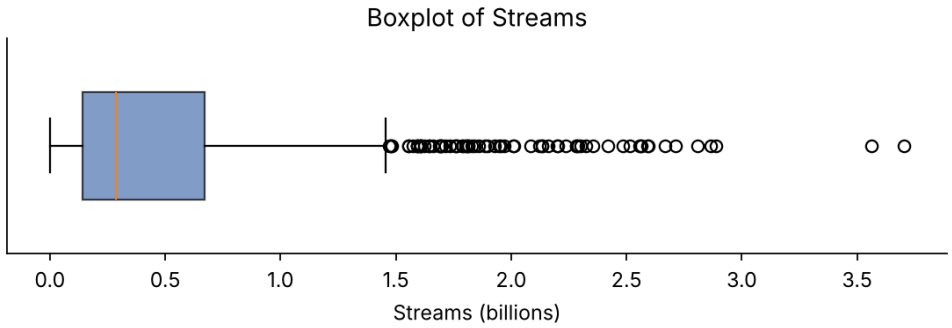


Figure 2: Boxplot of streams. There are 74 outliers above about 1.47 billion streams. They are real hit songs like Blinding Lights, so they were kept.

Q1: Which artist has the most songs in the list?

Answer: Taylor Swift has the most songs (34), then The Weeknd (21), and Bad Bunny and SZA (19 each).

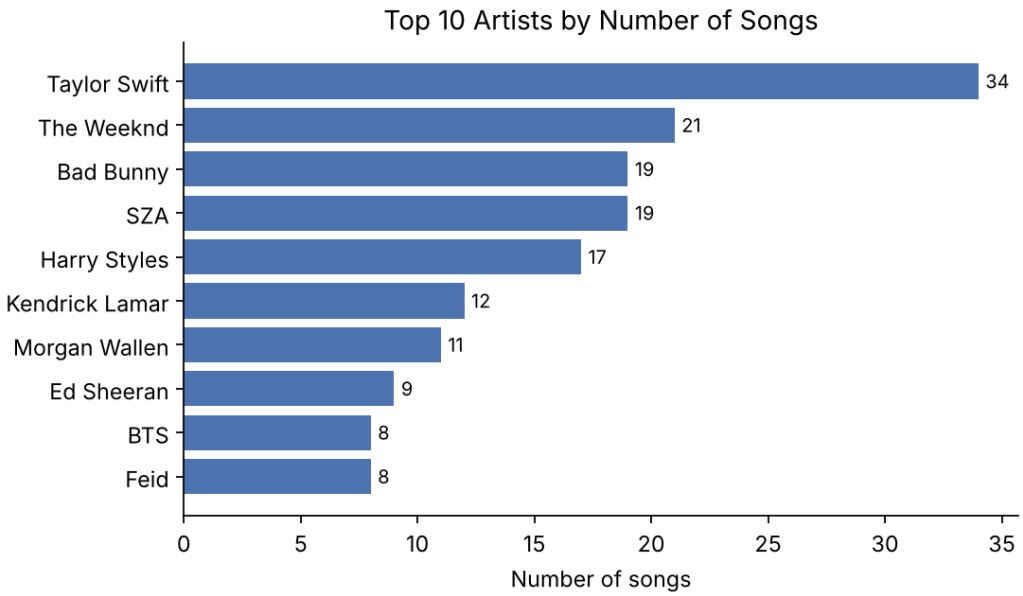


Figure 3: Top 10 artists by number of songs.

Q4: Are newer songs more common in the list?

Answer: Yes. About 77% of the songs were released in 2020 or later, and 2022 has the most songs (399).

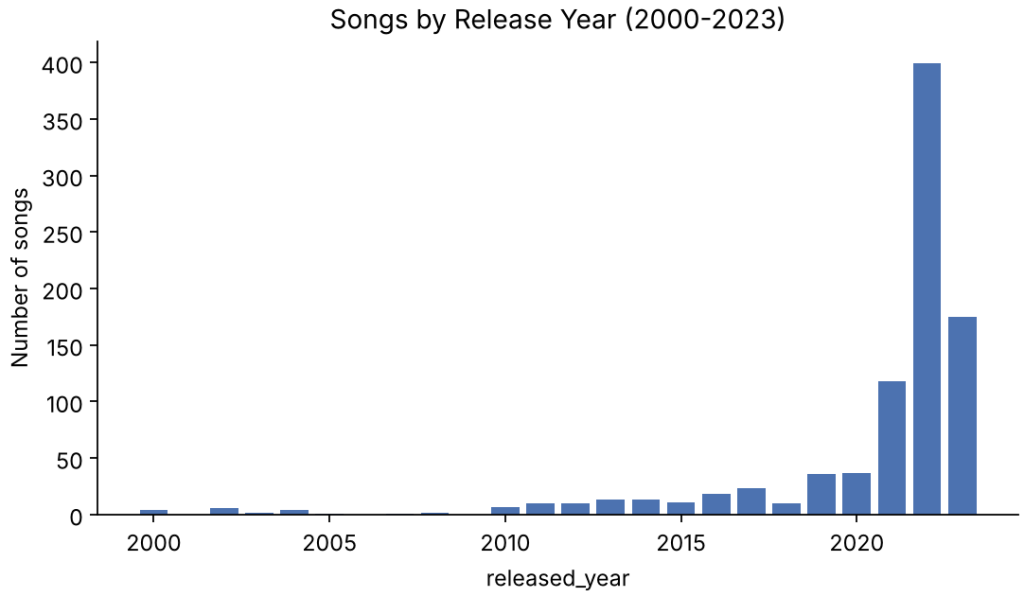


Figure 4: Number of songs by release year (2000-2023).

Q5: Are there more Major or Minor songs?

Answer: Major songs are more common (546) than Minor songs (402). Major songs also have a little more streams on average (535 million vs 485 million).

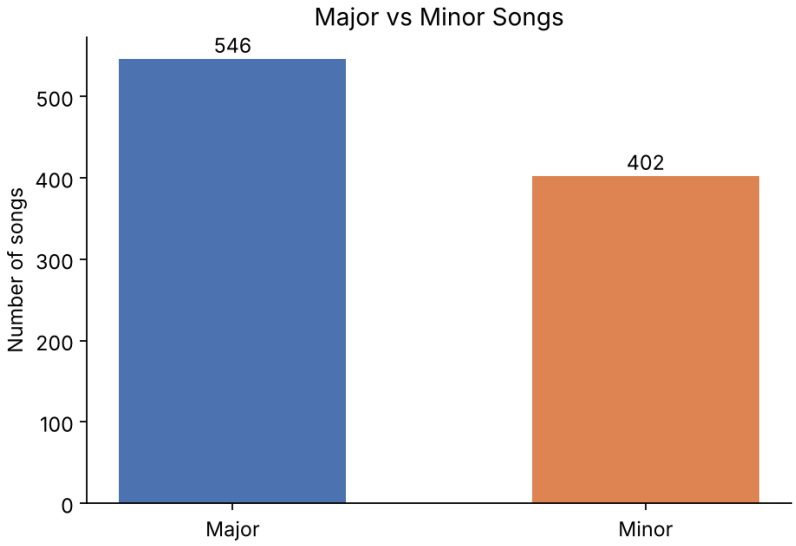


Figure 5: Number of Major and Minor songs.

Bivariate & Multivariate Analysis

Q2: What are the most streamed songs?

Answer: Blinding Lights by The Weeknd is the most streamed song (3.70 billion), then Shape of You by Ed Sheeran (3.56 billion). All of the top 10 songs were released between 2016 and 2021, so older songs had more time to collect streams.

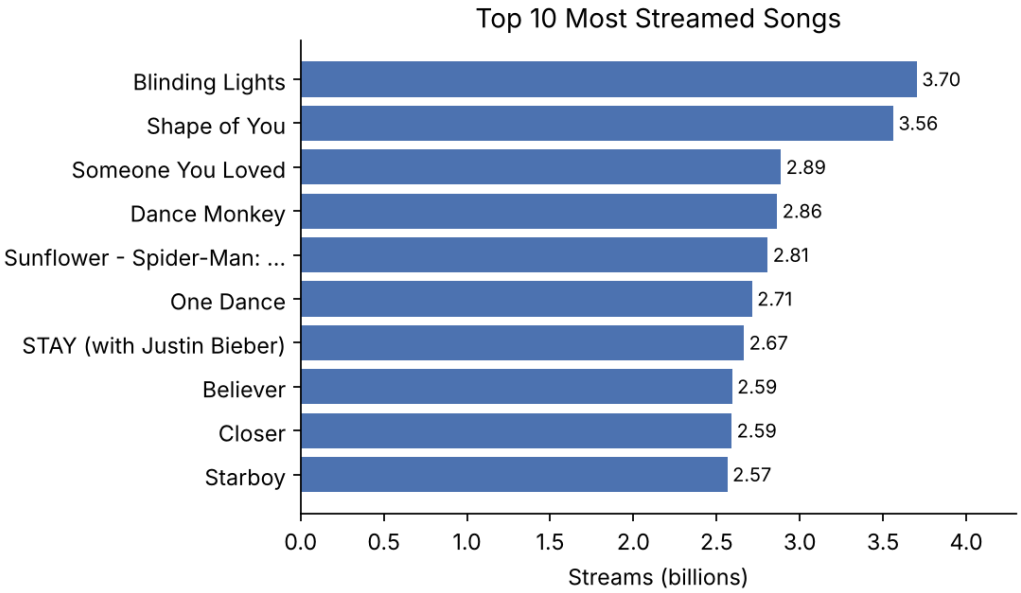


Figure 6: Top 10 most streamed songs.

Q3: Are danceable songs streamed more?

Answer: No. There is no clear relation between danceability and streams (correlation -0.11).

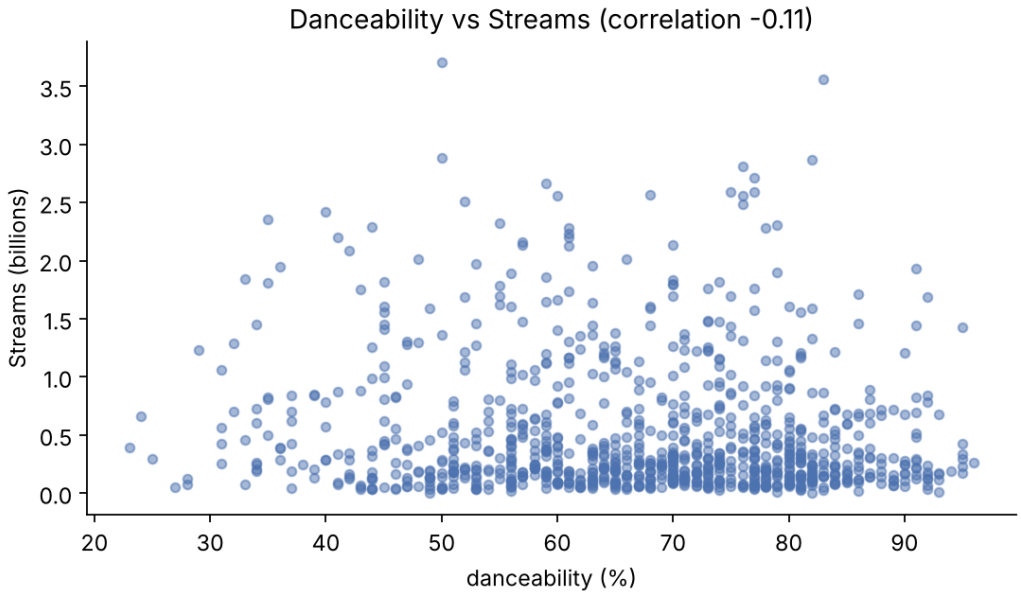


Figure 7: Scatter plot of danceability vs streams.

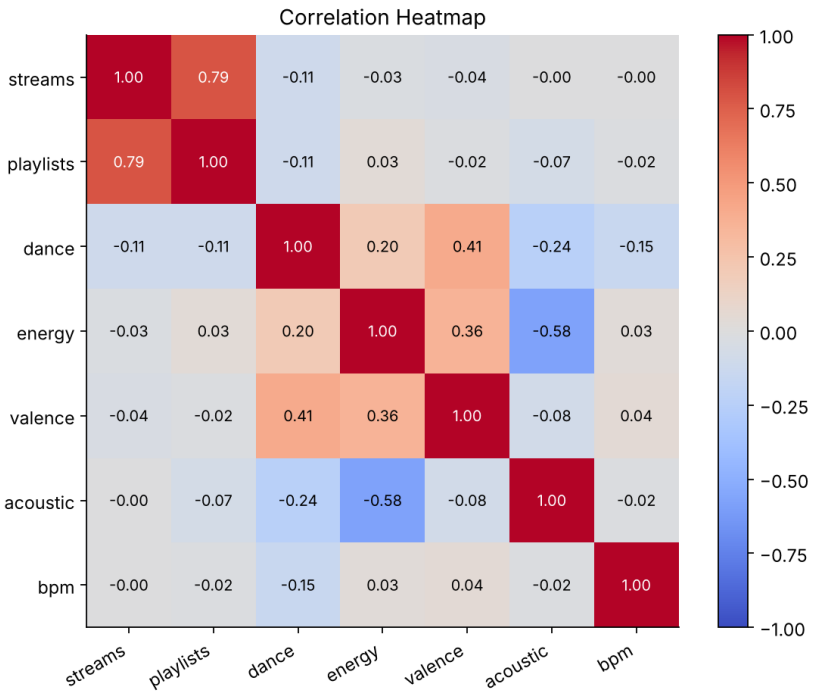


Figure 8: Correlation heatmap. Streams is strongly related to the number of Spotify playlists (0.79). The music features have almost no relation with streams. Energy and acousticness are negatively related (-0.58).

5. Dashboard Overview

An interactive dashboard was built with Streamlit (app.py). It reads clean_data.csv.

Dashboard link: [Add the Streamlit link here]

Sidebar

- Dataset description.
- **Dropdown** to choose the mode (All, Major or Minor).
- **Slider** to choose the release year range.
- **Multiselect** to choose artists (top 20 artists).

Main Page

- Data preview of the filtered songs.
- Summary statistics: number of songs, total streams, average danceability and describe().
- Interactive charts: top artists, top songs, songs by release year, danceability vs streams.
- Insights section.

[Add a screenshot of the dashboard here]

6. Insights & Conclusion

Main Insights

- Taylor Swift has the most songs in the list (34).
- Blinding Lights by The Weeknd is the most streamed song (3.70 billion streams).
- Most songs in the list are new: about 77% were released in 2020 or later.
- Major songs are more common than Minor songs.
- Music features (danceability, energy, valence, bpm) do not affect the number of streams.
- Songs in more Spotify playlists have more streams (correlation 0.79).

What I Learned

- How to find numbers saved as text and convert them.
- How to decide between filling and dropping missing values.
- How to choose the right chart for each question.
- How to build a Streamlit dashboard with filters.

Limitations of the Dataset

- The data is only for 2023, so it does not show changes over the years.
- Older songs had more time to collect streams, so comparing streams between old and new songs is not fair.
- The artist column mixes solo songs and songs with more than one artist.
- 95 songs have an unknown key.

Recommendations for Future Analysis

- Use data from more years to compare trends.
- Compare streams per day since release instead of total streams.
- Split songs with more than one artist to count each artist.
- Add the music genre to compare genres.

7. References

- Dataset: Most Streamed Spotify Songs 2023, Kaggle. https://www.kaggle.com/datasets/nelgiriyewithana/top-spotify-songs-2023
- GitHub repository: [Add the GitHub link here]
- Streamlit dashboard: [Add the Streamlit link here]
- Pandas documentation: https://pandas.pydata.org/docs/
- Seaborn documentation: https://seaborn.pydata.org/
- Streamlit documentation: https://docs.streamlit.io/